Kinematic and Singularity Analysis Report with Explicit Matrix Operations

Automated Generation System

June 9, 2026

1 Mathematical Formulation

The closed-loop spatial kinematic constraint is given by:

$$T_1(q_1)T_2(q_2)T_3(q_3)T_4(q_4)T_5(q_5)T_6(q_6)T_7(q_7) = I \tag{1}$$

The velocity constraint yields $J_p\dot{\mathbf{q}}_p = -J_a\dot{q}_1$, where J_p is the 6×6 passive Jacobian matrix obtained by removing the column corresponding to the driving joint J_1.

2 Example Matrix Calculation at $q_1 = 0°$

The solved full joint configuration (in radians) is:

$$\mathbf{q} = \begin{bmatrix} 0.0000 \\ 0.0010 \\ -0.0000 \\ -0.0020 \\ 0.0000 \\ -0.0010 \\ -0.0000 \end{bmatrix} \tag{2}$$

Based on this configuration, the full 6×7 Jacobian matrix J is evaluated as:

By extracting the columns corresponding to the passive joints, we form the passive Jacobian $J_p \in R^{6\times6}$:

$$J_p = \begin{bmatrix} -0.000 & 0.000 & 2.500 & -0.003 & -2.500 & 0.000 \\ -1.000 & -0.000 & 3.000 & -0.000 & 2.000 & 0.000 \\ 0.000 & -3.000 & 0.003 & -0.500 & 0.003 & -2.000 \\ 0.000 & 0.000 & 0.001 & 0.000 & 0.001 & -0.000 \\ 0.000 & -1.000 & -0.000 & -1.000 & 0.000 & 1.000 \\ 1.000 & 0.000 & -1.000 & 0.000 & 1.000 & 0.000 \end{bmatrix} \tag{3}$$

Applying Singular Value Decomposition (SVD) on $J_p = U\Sigma V^T$, the decomposition yields U:

$$U = \begin{bmatrix} -0.667 & -0.004 & -0.641 & 0.000 & 0.380 & 0.000 \\ -0.671 & 0.001 & 0.738 & -0.000 & 0.068 & 0.000 \\ -0.002 & 0.990 & -0.004 & 0.142 & -0.000 & 0.000 \\ -0.000 & 0.000 & 0.000 & 0.000 & 0.000 & -1.000 \\ -0.000 & 0.142 & -0.001 & -0.990 & -0.000 & -0.000 \\ 0.324 & 0.002 & 0.210 & -0.000 & 0.923 & 0.000 \end{bmatrix} \tag{4}$$

The singular values matrix Σ (diagonal):

$$\Sigma = \begin{bmatrix} 4.177 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 3.669 & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & 3.355 & 0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 1.669 & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.892 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \end{bmatrix} \tag{5}$$

And the transposed right singular vectors matrix V^T:

$$V^T = \begin{bmatrix} 0.238 & 0.002 & -0.959 & 0.001 & 0.155 & 0.001 \\ 0.000 & -0.848 & -0.001 & -0.174 & 0.004 & -0.501 \\ -0.157 & 0.003 & 0.119 & 0.001 & 0.980 & 0.002 \\ -0.000 & 0.338 & 0.000 & 0.551 & -0.000 & -0.763 \\ 0.958 & 0.001 & 0.258 & -0.001 & 0.123 & -0.000 \\ 0.001 & -0.408 & 0.000 & 0.816 & -0.000 & 0.408 \end{bmatrix} \tag{6}$$

3 Analysis Results & Curve Fitting

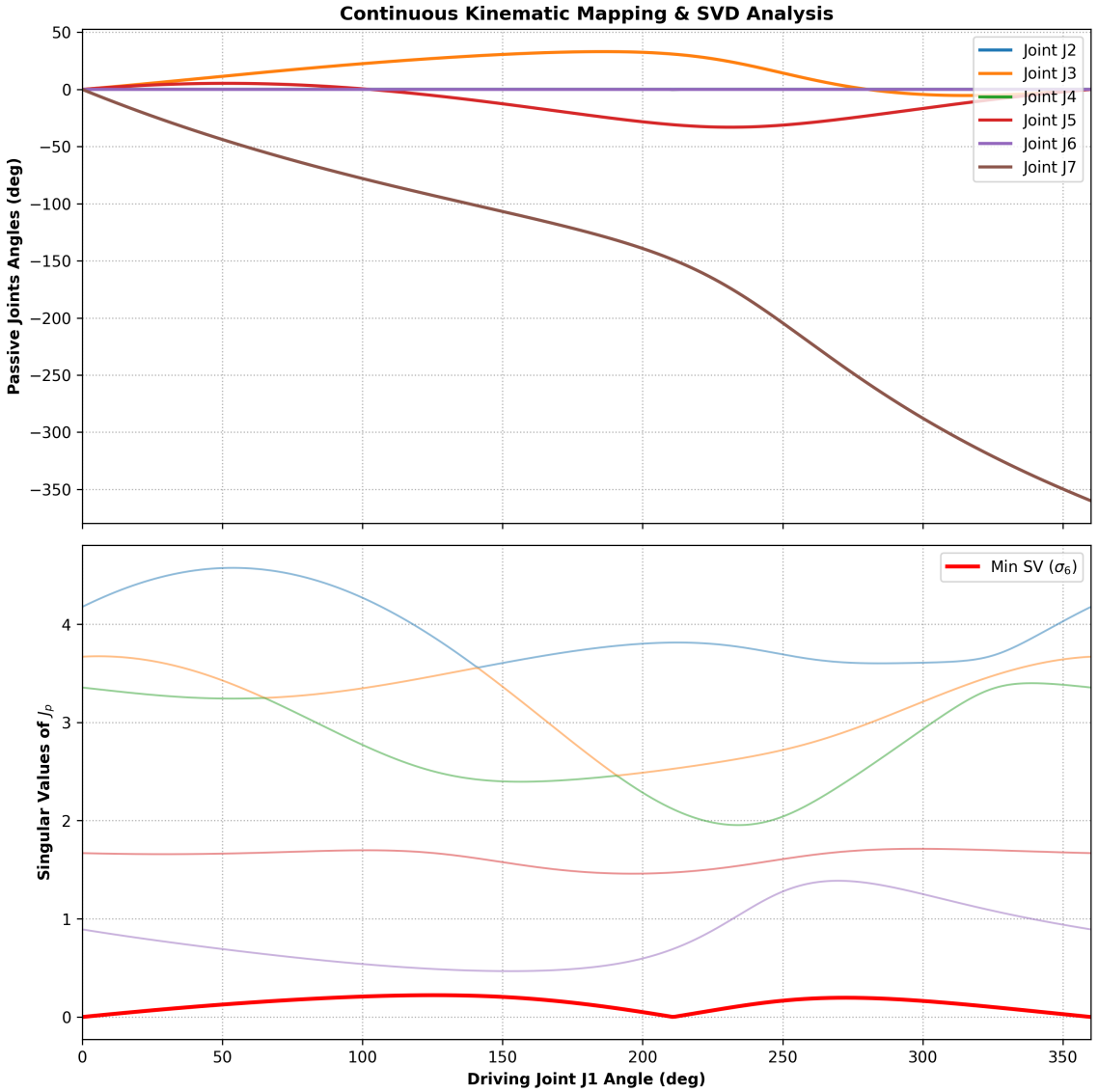


Figure 1: Kinematic mapping and singular value evolution driven by J_1.

The empirical curve fitting formulas for the passive joints are derived as follows:

- **Joint J2:**
  $Q_{J2}(x) = +0.000 - 0.001\cos(x) - 0.000\sin(x) + 0.000\cos(2x) + 0.001\sin(2x) - 0.000\cos(3x) - 0.001\sin(3x) - 0.000\cos(4x) + 0.001\sin(4x) + 0.000\cos(5x) - 0.000\sin(5x)$
- **Joint J3:**
  $Q_{J3}(x) = +14.741 - 17.158\cos(x) + 7.200\sin(x) + 2.094\cos(2x) + 3.424\sin(2x) + 0.619\cos(3x) - 0.949\sin(3x) - 0.366\cos(4x) - 0.023\sin(4x) + 0.062\cos(5x) + 0.112\sin(5x)$
- **Joint J4:**
  $Q_{J4}(x) = -0.000 + 0.001\cos(x) + 0.001\sin(x) - 0.000\cos(2x) - 0.001\sin(2x) + 0.000\cos(3x) + 0.001\sin(3x) + 0.001\cos(4x) - 0.001\sin(4x) - 0.001\cos(5x) + 0.000\sin(5x)$

- **Joint J5:**
  $Q_{J5}(x) = -11.728 + 11.633\cos(x) + 14.541\sin(x) + 0.432\cos(2x) - 1.922\sin(2x) - 0.460\cos(3x) + 0.224\sin(3x) + 0.143\cos(4x) + 0.068\sin(4x) - 0.013\cos(5x) - 0.053\sin(5x)$

- **Joint J6:**
  $Q_{J6}(x) = -0.000 + 0.001\cos(x) + 0.000\sin(x) - 0.000\cos(2x) - 0.001\sin(2x) + 0.000\cos(3x) + 0.001\sin(3x) + 0.000\cos(4x) - 0.001\sin(4x) - 0.000\cos(5x) + 0.000\sin(5x)$

- **Joint J7:**
  Model 1: 7th-order Polynomial
  $Q_{J7\text{-Poly}}(x) = -6.048020e-02x^7 + 1.278333e+00x^6 - 1.030244e+01x^5 + 3.982107e+01x^4 - 7.859981e+01x^3 + 8.140858e+01x^2 - 8.419649e+01x + 2.276139e+00$

  Model 2: 5th-order Fourier Series with Linear Trend
  $Q_{J7\text{-Fourier}}(x) = +26.400 - 57.274x - 28.791\cos(x) - 7.297\sin(x) + 1.662\cos(2x) + 5.368\sin(2x) + 1.079\cos(3x) - 1.158\sin(3x) - 0.509\cos(4x) - 0.079\sin(4x) + 0.074\cos(5x) + 0.174\sin(5x)$